

Good afternoon, everyone.

Today, I would like to introduce the process of an airplane taking off.

To begin with, once all passengers have boarded the aircraft, the cabin crew checks whether everyone is seated and whether the luggage has been properly stored. After this step is completed, the aircraft doors are closed, and passengers are asked to fasten their seat belts.

Subsequently, the aircraft is pushed back from the gate by a special vehicle called a pushback tug. During this stage, the engines are started by the pilots, and the engine sound gradually becomes louder.

Once the pushback has been completed, the aircraft begins to taxi toward the runway. This means that the airplane moves slowly on the ground. Meanwhile, the pilots communicate with the control tower and wait for further instructions.

Before the aircraft enters the runway, permission must be given by the control tower. As soon as takeoff clearance is received, the aircraft moves onto the runway and lines up with the centerline.

Following that, engine power is increased, and the aircraft starts to accelerate rapidly. During this stage, there is an important speed called V_1 . V_1 is not a fixed speed, but for many passenger aircraft, it is usually around 130 to 160 knots, which is about 240 to 300 kilometers per hour. Once the aircraft passes V_1 , the takeoff usually has to continue even if a problem occurs, because there may not be enough runway left to stop safely. When the required speed is reached, the nose of the airplane is lifted first. Then, the main wheels leave the ground, and the aircraft finally takes off.

After the aircraft has left the ground, the landing gear is retracted, and the airplane continues to climb. Eventually, it reaches a safe altitude and continues toward its cruising altitude.

These are the general steps for an airplane to take off. Thank you for listening.